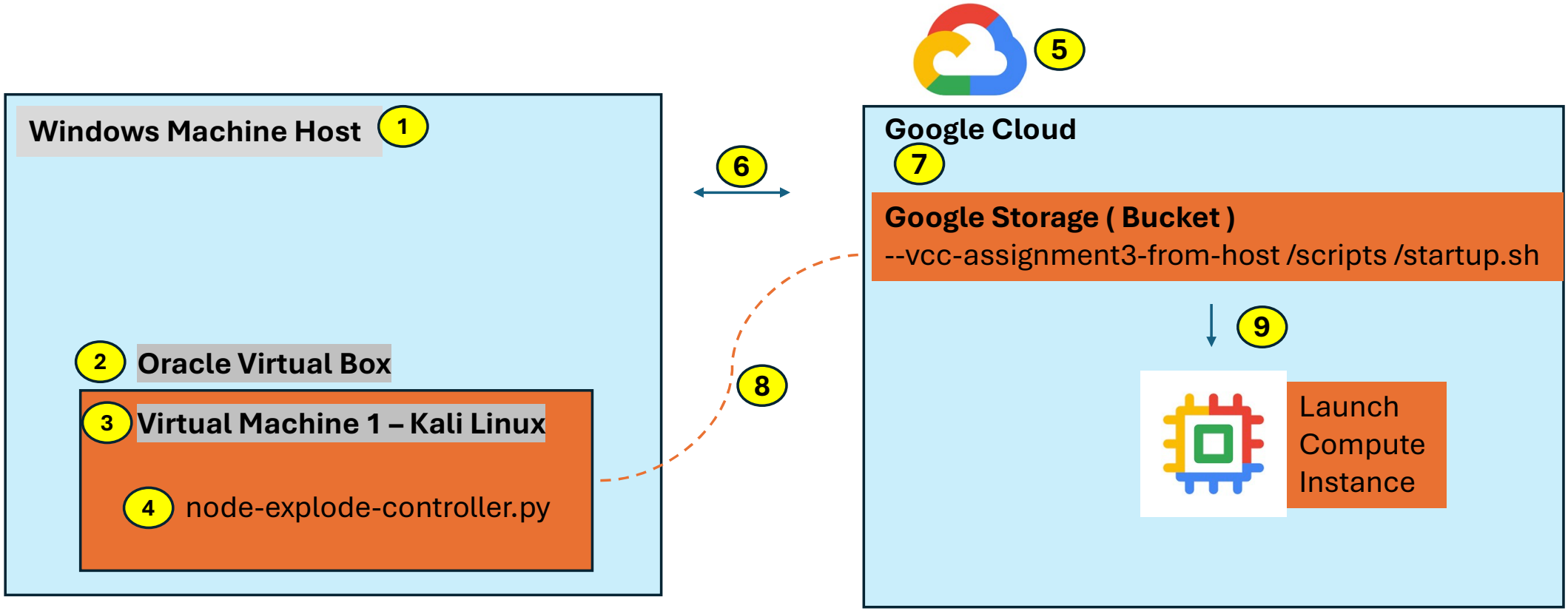
Objective:
Create a local VM and implement a mechanism to monitor resource usage. Configure it to auto-scale to a public cloud (e.g., GCP, AWS, or Azure) when resource usage exceeds 75%.

1. Host Machine
2. Oracle Virtual box installed in local machine
3. Virtual Machine 1 (Kali Linux) installed on Virtual Box
4. Create script called node-explode-controller.py
5. Configure Google Cloud account
6. Enable access between virtual machine and google cloud
7. Create bucket and create startup.sh script
8. In the virtual machine, execute note-explode-controller.py script
   In the virtual machine, launch another terminal and execute stress script
9. New compute instance would launch within google cloud



1. **Power on the VM (Kali Linux) installed within Oracle Virtual Box**
2. **Log into the machine with credentials.**

Update kali version

```
sudo apt update && sudo apt -y full-upgrade
sudo reboot
```

3. **Installing monitoring and load simulation tools in kali**

```
sudo apt install -y htop sysstat iotop python3-pip
sudo systemctl enable --now sysstat
```

Install stress tools:

```
sudo apt install -y stress-ng
```

4. **Install and configure Google Cloud CLI (gcloud) on Kali**

```
sudo apt install -y apt-transport-https
sudo apt install -y ca-certificates
sudo apt install -y gnupg
sudo apt install -y curl
curl -sSL https://packages.cloud.google.com/apt/doc/apt-key.gpg
sudo gpg --dearmor -o /usr/share/keyrings/cloud.google.gpg
echo "deb [signed-by=/usr/share/keyrings/cloud.google.gpg] https://packages.cloud.google.com/apt cloud-skd main" | sudo tee /etc/apt/source.list.d/google-cloud-sdk.list
sudo apt update
sudo apt install -y google-cloud-cli
gcloud --version
```

5. **Setting up Google Account and project.**

gcloud auth login

Login to google account and setup project

gcloud config set project vcc-assignment-3

gcloud config set compute/region asia-south1

gcloud config set compute/zone asia-south1-a

6. Create startup.sh script (Refer attached file in github)

startup.sh

7. Create load monitoring script called node-explode-controller.py

8. In kali Linux

open 1 terminal / bash and execute the node-explode-controller.py script.

This is the monitoring script that will evaluate the machine performance and issue command to launch node if the performance exceeds 75

node-explode-controller.py

command to execute:

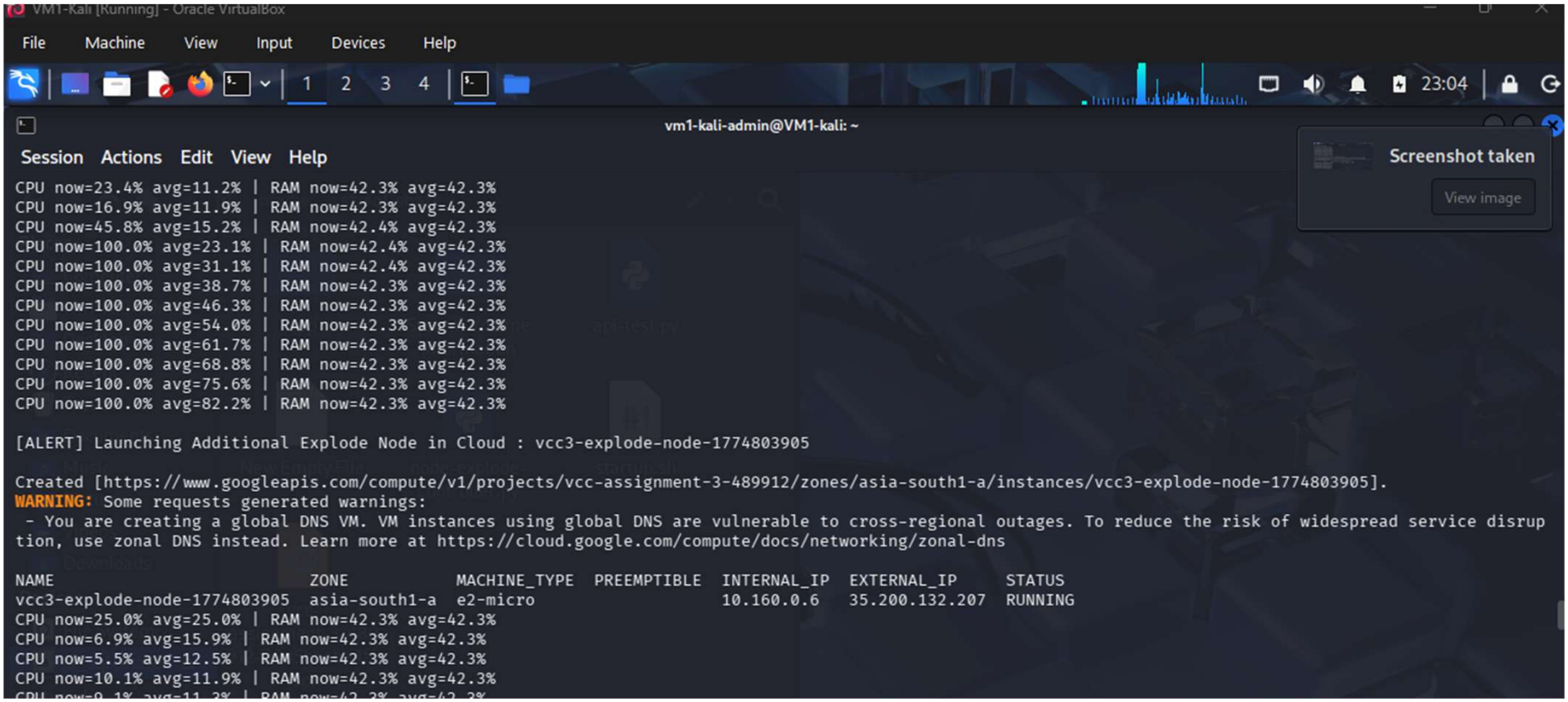
python3 node-explode-controller.py

8. In another terminal within kali Linux issue the stress command.

stress-ng --cpu 2 --cpu-load 85 --timeout 120s

GitHub Repo Link :

https://github.com/shyaamkr/Virtualization-Cloud-Computing---AutoScale-To-Google-Cloud-From-Local

Appendix:

Screenshot of launching node in google cloud once the threshold exceeds 75%